

Headline part VIII: the dressed normal moment integral

Claude, with Daniel Murfet

2026-09-07

Abstract

Paper-facing wrappers for the continuous-amplitude milestone (units 184–186). For the dressed normal moment $I_\eta(N) = \int_{(0,1]^{m+1}} \eta(x) x^h e^{-\beta N x^{2k}} dx$ with continuous η : the face-supported leading functional spelled out on the Bochner integral (`headline_normal_moment_amplitude`); the constant-amplitude consistency check (`amplitudeCoeff_const`); the equal-ratio case, where the coefficient is $\eta(0)$ times the bare constant (`headline_normal_moment_amplitude_equal`); the face-vanishing case with zero coefficient (`amplitudeCoeff_eq_zero_of_face`); and the equivalence form under a nonzero coefficient (`headline_normal_moment_amplitude_equiv`). Zero sorry/axiom.

1 The things formalised

```
theorem volume_unitBox (d) : volume (unitBox d) = 1
theorem amplitudeCoeff_const (hk) (hmin) (c) :
  amplitudeCoeff h k l (fun _ => c) = c * monomialMixedConst h k l
theorem amplitudeCoeff_eq_zero_of_face (h : u, (faceProj h k l u) = 0) :
  amplitudeCoeff h k l = 0
theorem amplitudeCoeff_equal (hratio : i, ratioExp h k i = 1) () :
  amplitudeCoeff h k l = 0 * (Gamma l * ^(-1) / (d.factorial * i, 2 * k i))
theorem headline_normal_moment_amplitude ... () (h : Continuous) : Tendsto
  (fun N => ( x in unitBox (m + 1), x * (( i, x i ^ h i) * exp (-
    ( * N * i, x i ^ (2 * k i))))
    / (N ^(-1) * log N ^ (|J| - 1))) atTop
  ( ((Gamma l * ^(-1) / (|J| - 1)! * i, if i = 1 then 1 / (2 * k i) else 1)
    * u in unitBox (m + 1), (fun i => if i = 1 then 0 else u i)
    * i, if i = 1 then 1 else u i ^ ((h i : ) - 2 * k i * l)))
theorem headline_normal_moment_amplitude_equal ... (hratio) : Tendsto (... / (N ^(-
1) * log N ^ m))
  atTop ( ( 0 * (Gamma l * ^(-1) / (m.factorial * i, 2 * k i))))
theorem headline_normal_moment_amplitude_equiv ... (hc : amplitudeCoeff h k l 0) :
  (fun N => ...) ~[atTop] fun N => amplitudeCoeff h k l * N ^ (-
1) * log N ^ (|J| - 1)
```

2 Paper-facing meaning

This closes the deterministic one-block programme at the level Astra #16 called “one normal block complete”: arbitrary mixed ratios, continuous amplitudes, an explicit face-supported leading functional, cutoff scaling, and an explicit treatment of a possibly vanishing leading coefficient (the asymptotic equivalence is asserted only when the coefficient is nonzero). Origin evaluation of the amplitude is verified to be correct exactly in the equal-ratio case.

3 Lean notes

- `setIntegral_const` now returns `.real s • c`; unfold with `measureReal_def` before using the `volume` value.
- `volume (unitBox d) = 1` is `volume_pi_pi` plus `Real.volume_Ioc`.
- A `congrArg` around `setIntegral_congr_fun` does not typecheck (the latter already produces the integral equality); use it directly.